

Multiple Linear Regression – Multi-Channel Marketing Analysis

Introduction

This project analyzes the relationship between multiple marketing channels and Sales using Multiple Linear Regression.

The analysis includes:

Exploratory Data Analysis (EDA) Categorical variable encoding Multicollinearity testing using VIF Regression modeling with statsmodels Residual diagnostics Business interpretation of coefficients

The objective is to determine which marketing channels significantly influence Sales and provide data-driven recommendations for marketing budget allocation.

1. Import Libraries

```
In [1]: import pandas as pd
import numpy as np

import matplotlib.pyplot as plt
import seaborn as sns

import statsmodels.api as sm
from statsmodels.stats.outliers_influence import variance_inflation_factor

from sklearn.preprocessing import LabelEncoder

from scipy import stats
```

2. Load Dataset

```
In [2]: df = pd.read_csv("marketing_sales_data.csv.csv")

df.head()
```

Out[2]:

	TV	Radio	Social Media	Influencer	Sales
0	Low	3.518070	2.293790	Micro	55.261284
1	Low	7.756876	2.572287	Mega	67.574904
2	High	20.348988	1.227180	Micro	272.250108
3	Medium	20.108487	2.728374	Mega	195.102176
4	High	31.653200	7.776978	Nano	273.960377

3. Initial Exploration

```
In [3]: print("Checking for information")
df.info()

print("\nChecking for missing Values")
df.isnull().sum()
```

```
Checking for information
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 572 entries, 0 to 571
Data columns (total 5 columns):
#   Column          Non-Null Count  Dtype
---  -
0   TV               572 non-null    object
1   Radio            572 non-null    float64
2   Social Media     572 non-null    float64
3   Influencer       572 non-null    object
4   Sales            572 non-null    float64
dtypes: float64(3), object(2)
memory usage: 22.5+ KB
```

Checking for missing Values

```
Out[3]: TV                0
Radio                0
Social Media        0
Influencer          0
Sales               0
dtype: int64
```

```
In [4]: print("\nChecking for Statistics")
df.describe()
```

Checking for Statistics

Out[4]:

	Radio	Social Media	Sales
count	572.000000	572.000000	572.000000
mean	17.520616	3.333803	189.296908
std	9.290933	2.238378	89.871581
min	0.109106	0.000031	33.509810
25%	10.699556	1.585549	118.718722
50%	17.149517	3.150111	184.005362
75%	24.606396	4.730408	264.500118
max	42.271579	11.403625	357.788195

We begin by inspecting the dataset structure, checking for missing values, and understanding the distribution of numerical variables.

4. Encode Categorical Variables

Since TV and Influencer are categorical:

```
In [5]: le_tv = LabelEncoder()
le_influencer = LabelEncoder()

df['TV'] = le_tv.fit_transform(df['TV'])
df['Influencer'] = le_influencer.fit_transform(df['Influencer'])

df.head()
```

Out[5]:

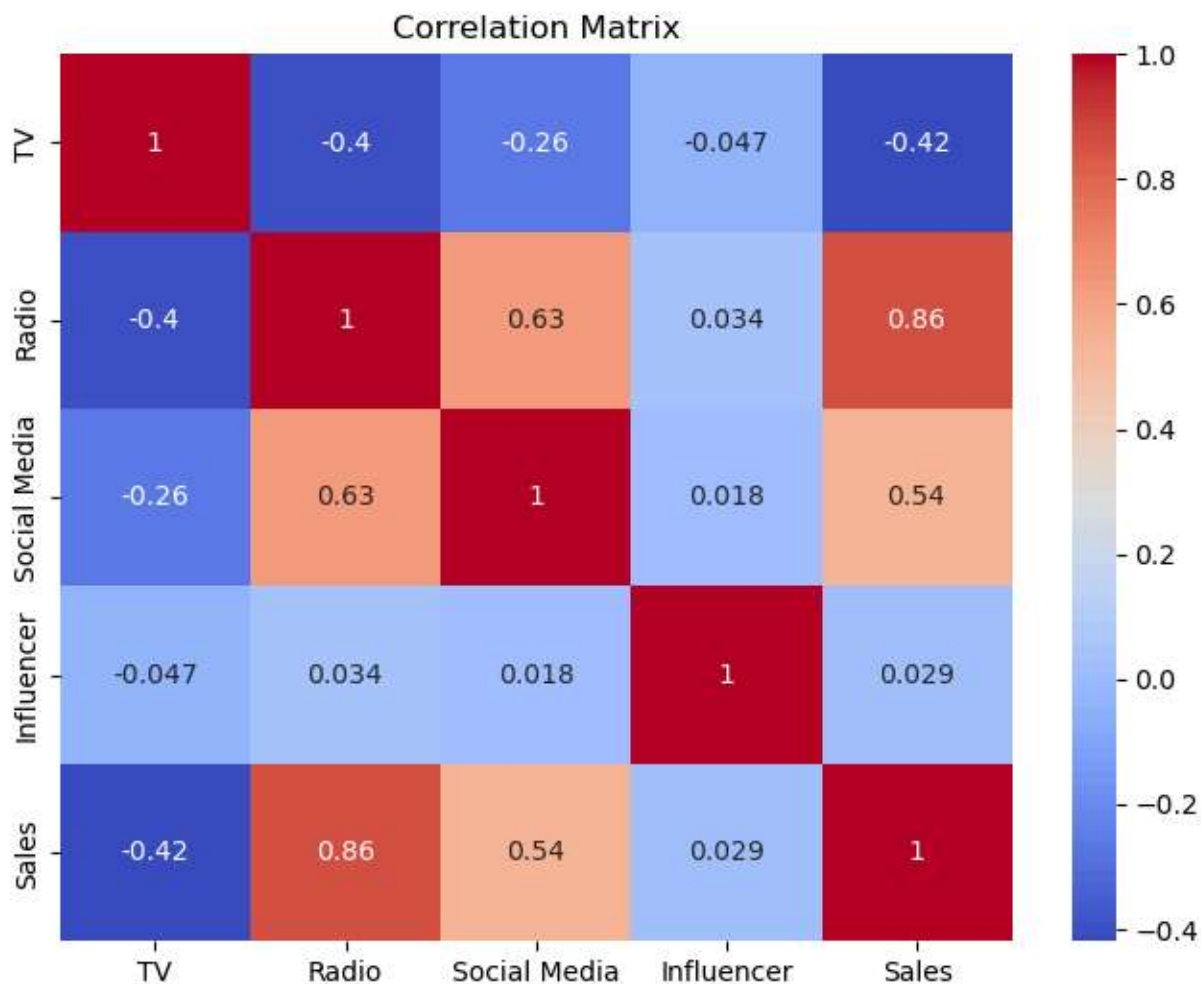
	TV	Radio	Social Media	Influencer	Sales
0	1	3.518070	2.293790	2	55.261284
1	1	7.756876	2.572287	1	67.574904
2	0	20.348988	1.227180	2	272.250108
3	2	20.108487	2.728374	1	195.102176
4	0	31.653200	7.776978	3	273.960377

Categorical variables were encoded into numerical representations to allow regression modeling.

5. Correlation Matrix

```
In [6]: plt.figure(figsize=(8,6))
sns.heatmap(df.corr(), annot=True, cmap='coolwarm')
```

```
plt.title("Correlation Matrix")
plt.show()
```



Interpretation:

Check relationships among predictors.

High correlations may indicate multicollinearity.

6. VIF (Variance Inflation Factor)

```
In [7]: X = df.drop("Sales", axis=1)

vif_data = pd.DataFrame()
vif_data["Feature"] = X.columns
vif_data["VIF"] = [variance_inflation_factor(X.values, i)
                    for i in range(X.shape[1])]

vif_data
```

Out[7]:

	Feature	VIF
0	TV	1.810678
1	Radio	6.005894
2	Social Media	5.258851
3	Influencer	2.517525

VIF values above 5 may indicate moderate multicollinearity, while values above 10 suggest severe multicollinearity.

7. Build Multiple Linear Regression Model

```
In [8]: X = df.drop("Sales", axis=1)
y = df["Sales"]

X = sm.add_constant(X)

model = sm.OLS(y, X).fit()

print(model.summary())
```

```

OLS Regression Results
=====
Dep. Variable:          Sales    R-squared:                0.744
Model:                  OLS      Adj. R-squared:           0.742
Method:                 Least Squares    F-statistic:            411.4
Date:                   Thu, 11 Jun 2026    Prob (F-statistic):      4.89e-166
Time:                   23:04:01    Log-Likelihood:          -2994.8
No. Observations:       572    AIC:                     6000.
Df Residuals:           567    BIC:                     6021.
Df Model:                4
Covariance Type:        nonrobust
=====
                    coef    std err          t      P>|t|      [0.025      0.975]
-----
const             61.9406      6.609      9.373      0.000      48.960      74.921
TV                -10.5653      2.593     -4.075      0.000     -15.658     -5.473
Radio              7.9363      0.278     28.513      0.000       7.390       8.483
Social Media       0.0214      1.099      0.019      0.984      -2.138       2.180
Influencer        -0.3001      1.720     -0.174      0.862      -3.679       3.078
=====
Omnibus:              0.291    Durbin-Watson:           1.848
Prob(Omnibus):         0.864    Jarque-Bera (JB):         0.244
Skew:                  0.050    Prob(JB):                 0.885
Kurtosis:              3.013    Cond. No.                 73.1
=====

```

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

8. Key Interpretation Section

Coefficient Interpretation

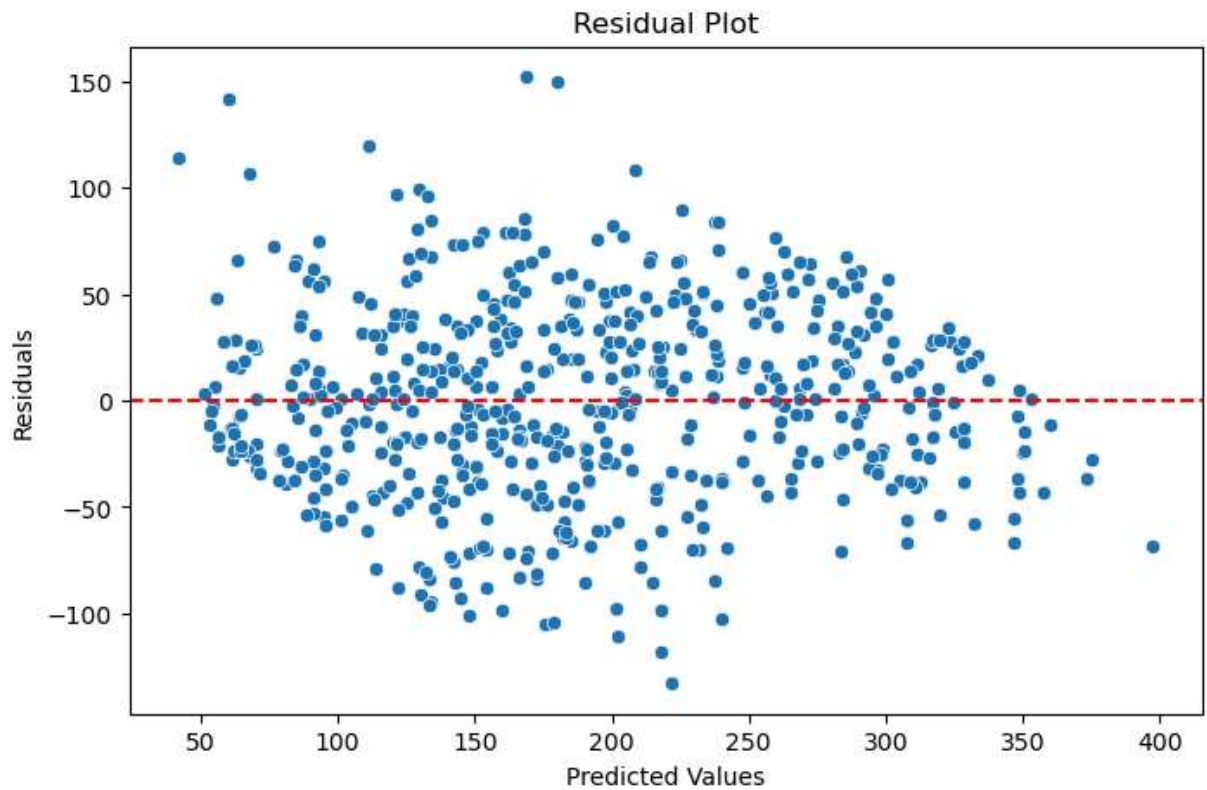
- Holding all other variables constant, a one-unit increase in Radio advertising spend is associated with an increase in predicted Sales equal to the corresponding regression coefficient.
- If the p-value of a variable is less than 0.05, the predictor is considered statistically significant.
- Adjusted R-squared explains the proportion of variation in Sales explained by the independent variables while accounting for model complexity.
- Variables with insignificant p-values may contribute little predictive power and can be considered for exclusion.

9. Residual Diagnostics

Residual Plot

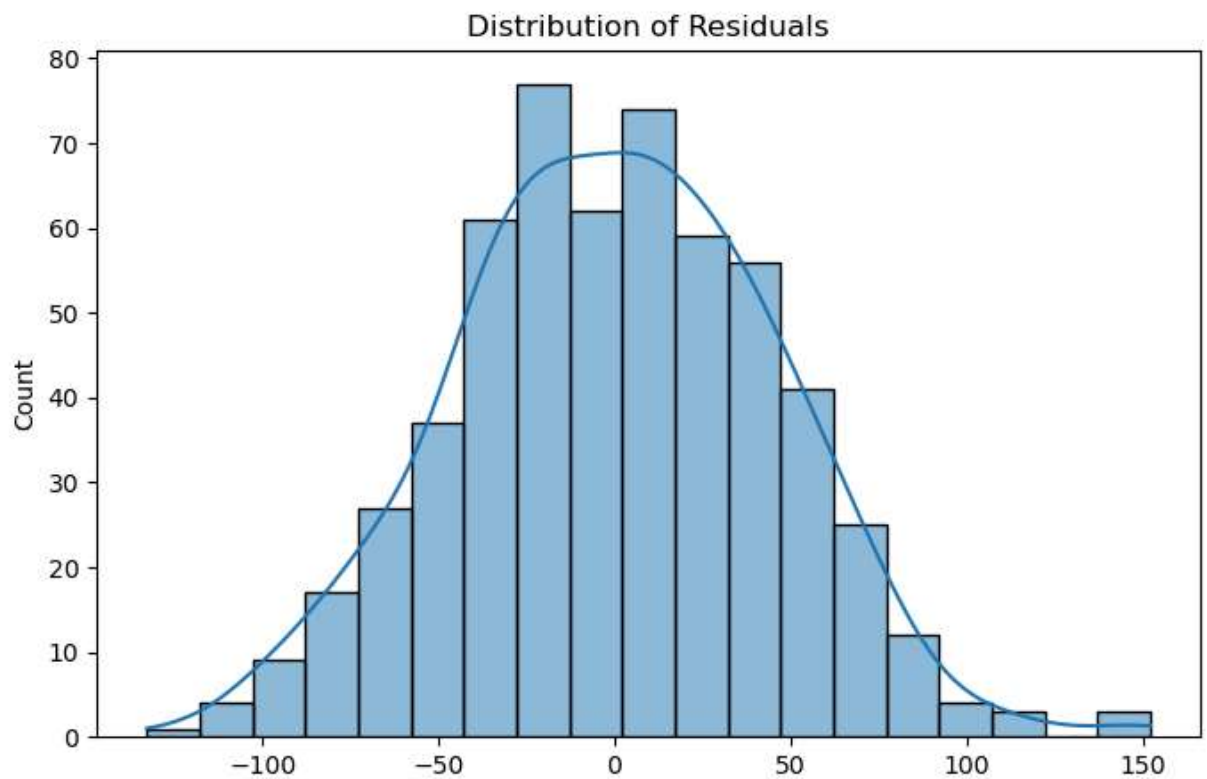
```
In [9]: predictions = model.predict(X)
        residuals = y - predictions

        plt.figure(figsize=(8,5))
        sns.scatterplot(x=predictions, y=residuals)
        plt.axhline(y=0, color='red', linestyle='--')
        plt.xlabel("Predicted Values")
        plt.ylabel("Residuals")
        plt.title("Residual Plot")
        plt.show()
```



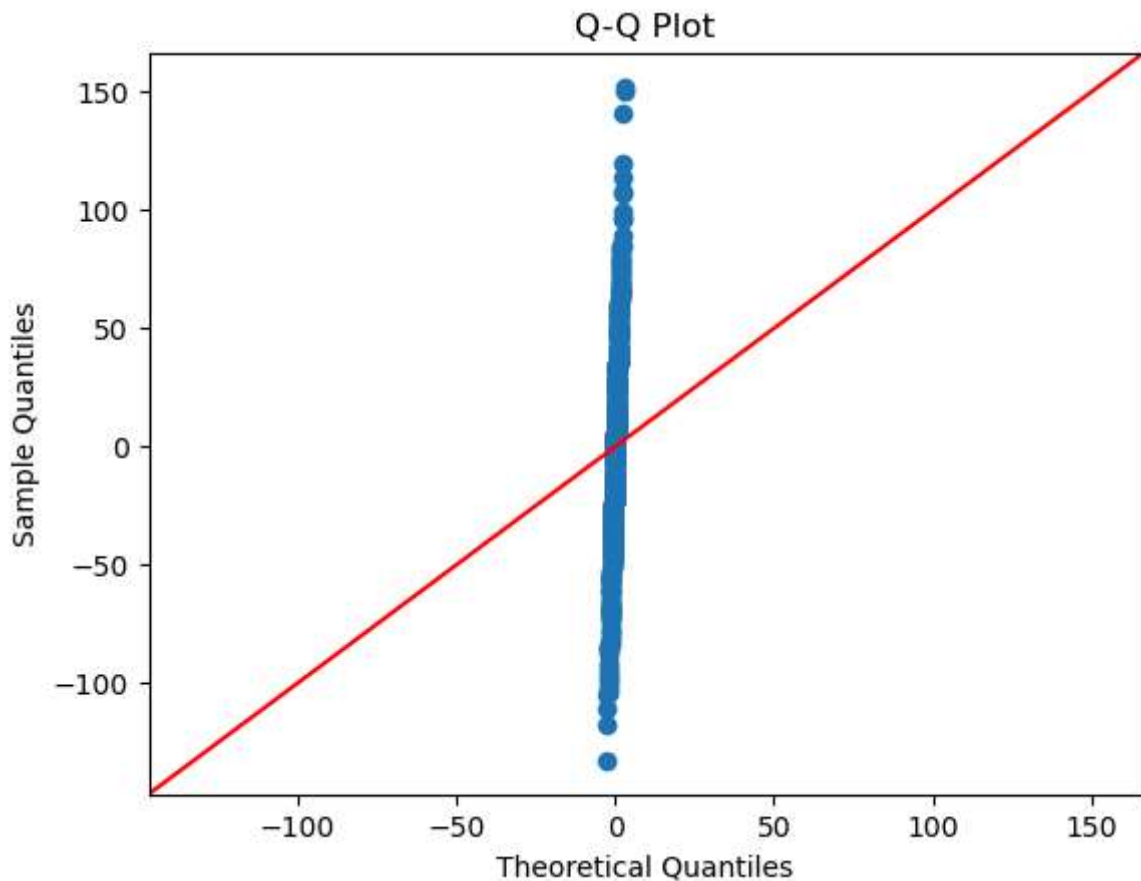
Histogram of Residuals

```
In [10]: plt.figure(figsize=(8,5))
sns.histplot(residuals, kde=True)
plt.title("Distribution of Residuals")
plt.show()
```



Q-Q Plot

```
In [11]: sm.qqplot(residuals, line='45')  
plt.title("Q-Q Plot")  
plt.show()
```



10. Business Recommendation

Business Recommendation

Based on the regression analysis, marketing channels with statistically significant positive coefficients should receive increased budget allocation because they contribute positively to Sales performance.

The Adjusted R-squared value indicates that the model explains a substantial proportion of Sales variability, suggesting good predictive performance.

Residual diagnostics show whether regression assumptions such as linearity, normality, and homoscedasticity are reasonably satisfied.

Management should prioritize investment in channels with:

- Significant p-values (< 0.05)

- Positive coefficient impact
- Strong contribution to overall model performance

Channels with weak statistical significance may require reduced spending or further investigation.

In []: